11.3.2 Water Potential (Ψ)

The concept of water potential was introduced in 1960 by **Slatyer** and **Taylor**. Water potential is potential energy of water in a system compared to pure water when both temperature and pressure are kept the same. It is also a measure of how freely water molecules can move in a particular environment or system. Water potential is denoted by the Greek symbol **Ψ** (psi) and measured in **Pascal** (Pa). At standard temperature, the water potential of pure water is **zero**. Addition of solute to pure water decreases the kinetic energy thereby decreasing the water potential. Comparatively a solution always has low water potential than pure water. In a group of cells with different water potential, a water potential gradient is generated. Water will move from higher water potential to lower water potential.

Water potential (**Ψ**) can be determined by,

1. Solute concentration or Solute potential (Ψ_S)
2. Pressure potential (Ψ_P)

By correlating two factors, water potential is written as,

$$\Psi_W = \Psi_S + \Psi_P$$

Water Potential = Solute potential + Pressure potential

1. Solute Potential (Ψ_S)

Solute potential, otherwise known as **osmotic potential** denotes the effect of dissolved solute on water potential. In pure water, the addition of solute reduces its free energy and lowers the water potential value from zero to negative. Thus the value of solute potential is always negative. In a solution at standard atmospheric pressure, water potential is always equal to solute potential ($\Psi_W = \Psi_S$).

2. Pressure Potential (Ψ_P)

Pressure potential is a mechanical force working against the effect of solute potential. Increased pressure potential will increase water potential and water enters cell and cells become **turgid**. This **positive hydrostatic pressure** within the cell is called **Turgor pressure**. Likewise, withdrawal of water from the cell decreases the water potential and the cell becomes **flaccid**.

3. Matric Potential (Ψ_M)

Matric potential represents the attraction between water and the **hydrating colloid or gel-like organic molecules in the cell wall** which is collectively termed as **matric potential**. Matric potential is also known as **imbibition pressure**. The matric potential is maximum (most negative value) in a dry material. **Example**: The swelling of soaked seeds in water.

11.3.3 Osmotic Pressure and Osmotic Potential

When a solution and its solvent (pure water) are separated by a semipermeable membrane, a pressure is developed in the solution, due to the presence of dissolved solutes. This is called **osmotic pressure** (**OP**). Osmotic pressure is increased with the increase of dissolved solutes in the solution. More concentrated solution (low **Ψ** or Hypertonic) has high osmotic pressure. Similarly, less concentrated solution (high **Ψ** or Hypotonic) has low osmotic pressure. The osmotic pressure of pure water is always **zero** and it increases with the increase of solute concentration. Thus osmotic pressure always has a positive value and it is represented as π.

Osmotic potential is defined as the ratio between the number of solute particles and the number of solvent particles in a solution. Osmotic potential and osmotic pressure are numerically equal. Osmotic potential has a negative value whereas on the other hand osmotic pressure has a positive value.

11.3.4 Turgor Pressure and Wall Pressure

When a plant cell is placed in pure water (hypotonic solution) the diffusion of water into the cell takes place by endosmosis. It creates a positive hydrostatic pressure on the rigid cell wall by the cell membrane. Henceforth the pressure exerted by the cell membrane towards the cell wall is **Turgor Pressure (TP).**

XIth_Botany_Chapter_11_EM.indd 238 05-01-2023 18:29:04